



Novel optical image encryption scheme based on fractional Mellin transform

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ARTICLE INFO

Article history:

Received 13 January 2011

Accepted 25 February 2011

Available online 12 March 2011

Keywords:

Fractional Mellin transform

Fractional Fourier transform

Amplitude encoding

Phase encoding

Image encryption

ABSTRACT

A novel nonlinear image encryption scheme is proposed by introducing the fractional Mellin transform (FrMT) into the field of image security. As a nonlinear transform, FrMT is employed to get rid of the potential insecurity of the optical image encryption system caused by the intrinsic object–image relationship between the plaintext and the ciphertext. Different annular domains of the original image are transformed by FrMTs of different orders, and then the outputs are further encrypted by comprehensively using fractional Fourier transform (FrFT), amplitude encoding and phase encoding. The keys of the encryption algorithm include the orders of the FrMTs, the radii of the FrMT domains, the order of the FrFT and the phases generated in the further encryption process, thus the key space is extremely large. An optoelectronic hybrid structure for the proposed scheme is also introduced. Numerical simulations demonstrate that the proposed algorithm is robust with noise immunity, sensitive to the keys, and outperforms the conventional linear encryption methods to counteract some attacks.

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1. Introduction

With the rapid development of the network multimedia, communication and propagation techniques, information security including image security is of serious concern. Image encryption is one of the effective approaches to ensure the security of the private image information. Various optical as well as digital techniques have been proposed for image encryption [1–12], including the well known double random phase encoding technique [1]. In this pioneer technique, two statistically independent random phase masks are used in the input and Fourier planes to encrypt the original image into a stationary white noise. Unnikrishnan extended this technique to the fractional Fourier domain [2], and then the FrFT is widely used in image encryption due to its good properties and feasibility with optical implementation [8–12]. In recent years, many transforms as new tools are transplanted to encrypt the private images, such as fractional cosine transform [13], multiple-parameter discrete fractional Fourier transform [14], fractional Hartley transform [15,16], and so on. Most of the above-mentioned transforms are linear transforms. As is well known, the linear encryption system is relatively weak in comparison to the nonlinear encryption system [17,18]. The functional relationship among plaintext, ciphertext and key is comparatively simple in a linear encryption system, which makes the encryption system potentially insecure against some common attacks such as chosen- and known-plaintext attacks. Especially, for a conventional optical image encryption, the potential insecurity is more obvious, since plaintext and ciphertext satisfy essentially linear object–image relation [19].

In this paper, we propose a nonlinear image encryption algorithm based on the fractional Mellin transform (FrMT). The FrMT is a nonlinear transform which helps to make the encryption system nonlinear. In accordance with the characteristics of the FrMT, different annular parts of the original image with center at the geometric center of the original image are transformed by the FrMTs of different orders. The outputs are several complex-valued images with the same sizes. Amplitude encoding and phase encoding based on fractional Fourier transform are used to encrypt these complex-valued images into one ciphertext. Comprehensive application of amplitude and phase encoding makes it more difficult to attack the encryption system, since some attacks [18,20] only work for the case of amplitude-coded plain images. Thus, the security of the encryption system is strengthened. The main keys of the proposed algorithm are some phases generated in the encryption process, and they are relevant to the original image. Moreover, the orders of the FrMTs, the outer radii of the FrMT domains and the order of the FrFT are keys optionally. Thus the key space of the proposed algorithm is very large. Simulation results demonstrate the feasibility, effectiveness, robustness and security of the proposed algorithm.

2. Basic theory

Fractional Fourier transform is a generalization of the conventional Fourier transform and is one of the most widely used tools in signal processing and optics. Conventionally, the α order FrFT [21,22] of input function $f(x)$ can be defined as follows:

$$F^\alpha\{f(x)\}(u) = \int_{-\infty}^{+\infty} K_\alpha(x, u) f(x) dx, \quad (1)$$

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where the transform kernel is given by

$$K_{\alpha}(x, u) = \begin{cases} A \exp \left[i\pi \left(x^2 \cot \phi - 2xu \csc \phi + u^2 \cot \phi \right) \right] & \text{if } \alpha \neq n\pi \\ \delta(x-u) & \text{if } \alpha = 2n\pi \\ \delta(x+u) & \text{if } \alpha = (2n+1)\pi \end{cases}, \quad (2)$$

and

$$A = \frac{\exp[-i(\pi \operatorname{sgn}(\phi) / 4 - \phi / 2)]}{\sqrt{|\sin \phi|}}, \quad (3)$$

where $\phi = \alpha\pi/2$ is the angle corresponding to the transform order along the x -axis. When $\alpha=1$, Eq. (1) reduces to the conventional Fourier transform. FrFT is linear and has the property that it is index additive. The extension of the definition to two-dimensional signals is straightforward and simple. The optical implementation [23] of the FrFT is finished by changing the distances and the focal length of the lenses performing the Fourier transform and its inverse.

Fractional Mellin transform is a generalization of the scale covariant transform and Mellin transform on the scale warped time–frequency plane. It also develops the conception of fractional integral transform. The definition of FrMT is inspired by fractional Fourier transform. The two-dimensional FrMT [24,25] of $f(x, y)$ with order (p_1, p_2) can be defined as

$$M^{(p_1, p_2)}(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) x^{-(2i\pi p_1 / \sin \Phi_1) - 1} \exp \left[\frac{i\pi}{\tan \Phi_1} (u^2 + \ln^2 x) \right] \\ \times y^{-(2i\pi p_2 / \sin \Phi_2) - 1} \exp \left[\frac{i\pi}{\tan \Phi_2} (v^2 + \ln^2 y) \right] dx dy \quad (4)$$

where $\Phi_1 = \frac{p_1 \cdot \pi}{2}$ and $\Phi_2 = \frac{p_2 \cdot \pi}{2}$. FrMT can be obtained by a change of variables of FrFT. In general case, FrMT with order (p_1, p_2) of an image is FrFT with the same order (p_1, p_2) of the image in its log-polar representation, which is shown as follows

$$M^{(p_1, p_2)}(u, v) = C \cdot \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\rho, \theta) \times \exp \left[-2\pi i \left(\frac{u\rho}{\sin \Phi_1} + \frac{v\theta}{\sin \Phi_2} \right) \right] \\ + \pi i \left(\frac{u^2 + \rho^2}{\tan \Phi_1} + \frac{v^2 + \theta^2}{\tan \Phi_2} \right) \right] d\rho d\theta \\ = F^{(p_1, p_2)}(f(\rho, \theta)) \quad (5)$$

where C is a constant and $\begin{cases} \rho = \ln \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \end{cases}$. The relationship between FrMT and FrFT inspires a fast and easy way to implement FrMT which is demonstrated in Fig. 1. The main problem for the implementation of FrMT is to realize the change of coordinates from (x, y) to (ρ, θ) . This problem can be solved in spatial coordinates by placing a hologram after the input field. Ref. [26] gives the set-up for

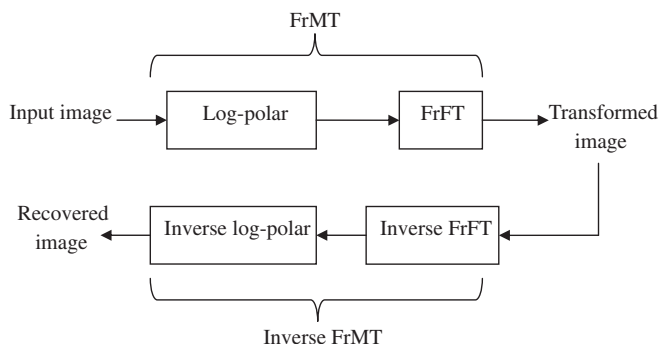


Fig. 1. FrMT and its inverse transformation.

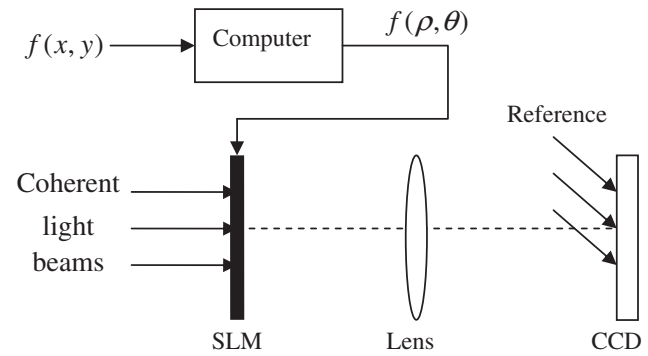


Fig. 2. Optoelectronic hybrid setup for FrMT.

the implementation of FrMT using computer-generated holograms. However, an easier method to transform the coordinate is using the optoelectronic hybrid system. As is shown in Fig. 2, image preprocessing of coordinate transformation is carried out through computer, and the output is then written to spatial light modulator to modulate coherent light beams. After that, the light beams pass through lens performing the FrFT, and the result of FrMT is obtained in the output plane which can be captured by CCD.

In Ref. [27], log-polar transformation is described in detail. Involving log-polar transformation process, FrMT is carried out in annular domain, so we have to set some parameters in advance, such as the center position of the annular domain in the original image (denoted as $(center_x, center_y)$), the radii of the innermost and outermost rings of the annular domain (denoted as r_{in}, r_{out}) and the number of discrete sampling points along distance axis and along angle axis (denoted as nr, nw). In theory, the position of any pixel in the original image can be set as $(center_x, center_y)$. r_{in} and r_{out} have to satisfy $1 \leq r_{in} \leq r_{out} \leq r_{max}$, where $r_{max} = \max(\sqrt{(x-center_x)^2 + (y-center_y)^2})$. In the log-polar sampling process, nr and nw can be vividly described as the number of rings and the number of wedges. They determine the size of transformed image and the quality of recovered image. The larger the values of nr and nw are, the better recovered image is, whereas the size of transformed image would be larger. So nr and nw should be comprehensively considered.

3. Image encryption and decryption algorithms based on fractional Mellin transform

Centered at geometric center of the original image $f(x, y)$, different annular domains of the image $f(x, y)$ are transformed by FrMTs of different orders. Assuming $f(x, y)$ consists of N annular subimages denoted as $f_i(x, y)$, $i = 1, 2, \dots, N$, an illustration when $N=4$ is shown in Fig. 3. $f_1(x, y)$ is of circle domain with radius r_1 ; $f_2(x, y)$ is of annular domain with inner radius r_1 and outer radius r_2 , and so on. $f_i(x, y)$, $i = 1, 2, \dots, N$ is transformed by p_i th order FrMT (assuming transform orders in both dimensions are identical), and we obtain transformed image $g_i(x, y)$, $i = 1, 2, \dots, N$, which is a complex-valued image. In order

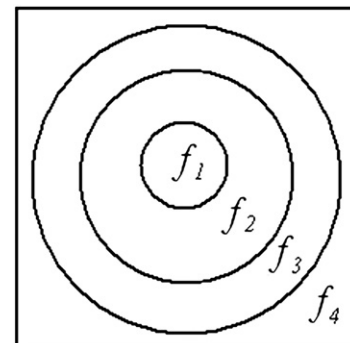


Fig. 3. $f(x, y)$ consists of four subimages.

to get the transformed images with the same sizes, one should make sure that the numbers of discrete sampling points along distance axis are same during N times of the FrMTs as stated above, and so are the numbers of sampling points along angle axis. However, in practice there is a negative effect of aliasing that is pixels near the edge of the original image cannot be recovered so well after the inverse FrMTs. To resolve this problem, when $f_i(x, y)$ is transformed by the FrMT, the radius of the innermost ring of the annular domain namely r_{in} is smaller than r_{i-1} . In this way, the edge part of the recovered $f_{i-1}(x, y)$ can be remedied by extra part of the recovered $f_i(x, y)$.

N transformed images are further encrypted into one encrypted image by the iterative process described as follows. We adopt amplitude encoding and phase encoding simultaneously. Amplitude of $g_i(x, y)$, $i = 1, 2, \dots, N$ is encoded into phase information and restricted into the range of phase distribution $[0, 2\pi]$, and phase of $g_i(x, y)$ is extracted as

$$A_i(x, y) = |g_i(x, y)|, \quad (6)$$

$$A'_i(x, y) = 2\pi \cdot A_i(x, y) / 360, \quad (7)$$

$$\Phi_i(x, y) = \text{Arg}[g_i(x, y)], \quad (8)$$

where $|\cdot|$ and $\text{Arg}[\cdot]$ represent extracting amplitude information and phase information, respectively. The iterative process is illustrated in Fig. 4. $\varphi_1(x, y)$ is encoded into amplitude information $C_1(x, y)$ and restricted into the range $[0, 255]$. Let $\xi_1(x, y) = A'_1(x, y)$, $B_1(x, y) = 0$, $\varphi_1(x, y) = 0$, $\psi_1(x, y) = 0$, $\theta_1(x, y) = 0$ and the iterative process begins with $i = 2$ as

$$B_i(x, y) \exp[i\varphi_i(x, y)] = F^q \{C_{i-1}(x, y) \cdot \exp[iA'_i(x, y)]\}, \quad (9)$$

where $B_i(x, y)$ and $\varphi_i(x, y)$ denote amplitude and phase, respectively, and $F^q\{\cdot\}$ represents fractional Fourier transform of order q (assuming transform orders in both dimensions are identical). The phase $\psi_i(x, y)$ is calculated as

$$\psi_i(x, y) = \xi_{i-1}(x, y) - A'_i(x, y), \quad (10)$$

which is regarded as the key of the proposed algorithm. And then $\varphi_i(x, y)$ is encrypted as

$$\theta_i(x, y) = \varphi_i(x, y) - \psi_i(x, y). \quad (11)$$

$\theta_i(x, y)$ is also the key of the proposed algorithm. $\Phi_i(x, y)$ combining with $\theta_i(x, y)$ is the phase information and $B_i(x, y)$ is the amplitude information. They are transformed by fractional Fourier transform,

$$C_i(x, y) \exp[i\xi_i(x, y)] = F^q \{B_i(x, y) \exp[i(\theta_i(x, y) + \Phi_i(x, y))]\}. \quad (12)$$

Substituting $C_{i-1}(x, y)$ with $C_i(x, y)$, $\xi_{i-1}(x, y)$ with $\xi_i(x, y)$ in Eqs. (9) and (10) respectively, and repeating the above process. After $N - 1$ iterations, we obtain $C(x, y) = C_N(x, y) \exp[i\xi_N(x, y)]$ which is the public data in this encryption algorithm. All keys involved in the

proposed encryption algorithm are the orders of the FrMTs, r_i , the orders of the FrFTs, phase θ_i and ψ_i .

Image decryption is an inverse process of encryption. The first step is to recover N subimages by inverse iterative process started from $i = N$. With correct key θ_i , $\Phi_i(x, y)$, $B_i(x, y)$ can be obtained as

$$\Phi_i(x, y) = \text{Arg}[F^{-q}\{C_i(x, y) \exp[i\xi_i(x, y)]\}] - \theta_i(x, y), \quad (13)$$

$$B_i(x, y) = F^{-q}\{C_i(x, y) \exp[i\xi_i(x, y)]\}. \quad (14)$$

And with correct key ψ_i , $A_i(x, y)$ is reconstructed by

$$A'_i(x, y) = \text{Arg}[F^{-q}\{B_i(x, y) \exp[i(\theta_i(x, y) + \psi_i(x, y))]\}], \quad (15)$$

$$A_i(x, y) = 360A'_i / (2\pi). \quad (16)$$

Meanwhile, $C_{i-1}(x, y)$ and $\xi_{i-1}(x, y)$ can be obtained.

$$C_{i-1}(x, y) = |F^{-q}\{B_i(x, y) \exp[i(\theta_i(x, y) + \psi_i(x, y))]\}|, \quad (17)$$

$$\xi_{i-1}(x, y) = \psi_i(x, y) + A'_i(x, y). \quad (18)$$

$C_{i-1}(x, y)$ and $\xi_{i-1}(x, y)$ should substitute for $C_i(x, y)$ and $\xi_i(x, y)$, respectively in Eqs. (13) and (14) to begin the next iteration. After $N - 1$ inverse iterations, $g_i(x, y)$ can be reconstructed as

$$g_i(x, y) = A_i(x, y) \exp[i\Phi_i(x, y)], \quad (19)$$

where $A_i(x, y)$ is amplitude form transformed from $\xi_i(x, y)$, and $\Phi_i(x, y)$ is phase form transformed from $C_i(x, y)$. $g_i(x, y)$ is transformed by the inverse FrMT of corresponding order and $f_i(x, y)$ can be obtained. By adding up all $f_i(x, y)$, the decrypted image can be obtained as

$$f(x, y) = \sum_{i=1}^N f_i(x, y) = \sum_{i=1}^N M^{(-p_i, -p_i)} \{g_i(x, y)\}. \quad (20)$$

4. Optical realization

A possible optical implementation based on the proposed encryption algorithm is shown in Fig. 5.

Different annular domains of the original image are transformed to log-polar coordinate by computer, and the output is written into the first spatial light modulator (SLM1). After the FrFT by using the first lens (Lens1), the subimages transformed by the FrMT are obtained on the output plane and received by the first CCD (CCD1). The output is fed into computer to start the iterative process. The second and third spatial light modulators (SLM2 and SLM3) can be used to carry out amplitude encoding and phase encoding, respectively. The second lens (Lens2) is used to carry out fractional Fourier transform. The result of each iteration is recorded by the second CCD (CCD2) and then transmitted to computer for further processing. The result of the encryption system can be recorded by digital holography [28,29].

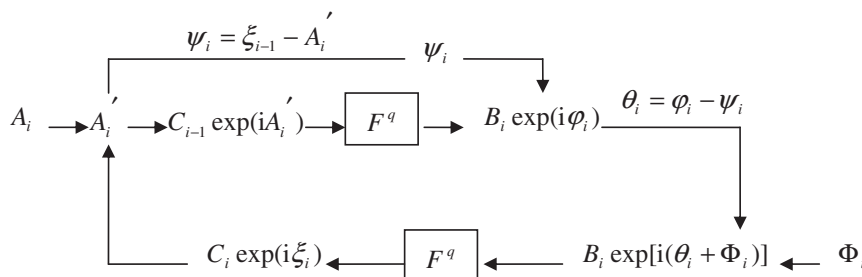


Fig. 4. The iterative process.

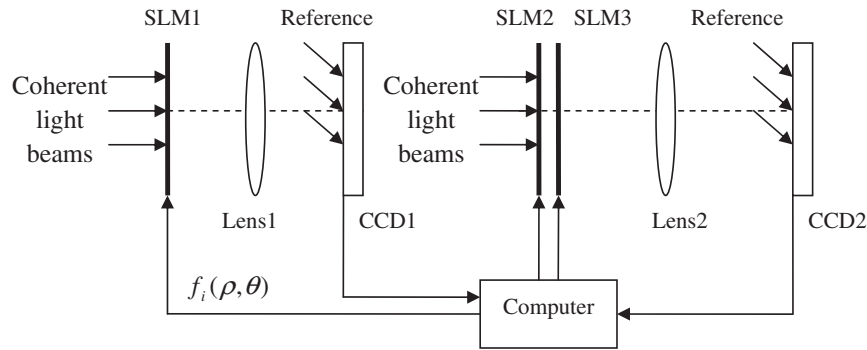


Fig. 5. Optoelectronic hybrid setup of the proposed encryption algorithm.

5. Simulation and discussion

The FrMT is fitted to operate on the image with odd length and odd width, so the gray image Lena with 255×255 pixels is used as a test image here, which is shown in Fig. 6a. We assume that the original image Lena consists of 5 annular subimages. For convenience, the transformation parameters of the rows and the columns are equal, and all of the FrMT orders are equal to 0.5. The FrMT center is the geometric center of the original image denoted as $(center_x, center_y) = (128, 128)$. To get decrypted image with good quality and not to increase the burden of image transmission too much, both nr and nw are set as 500. The outer radii of the annular domains are $r_1 = 50$, $r_2 = 85$, $r_3 = 135$, $r_4 = 160$, and $r_5 = 181$. All of the FrFT orders in the iterative process are equal to 0.5. The amplitude and phase distributions of the encrypted image are given in Fig. 6b and c, respectively. The decrypted image with correct keys is shown in Fig. 6d.

Fig. 7a–e are the decrypted images with incorrect keys. Fig. 7a is the decrypted image with incorrect inverse FrMT order $p_i = 0.55$, $i = 1, 2, \dots, 5$, while the other keys are correct. Fig. 7b gives the decrypted image with incorrect inverse FrFT order $q = 0.6$ while the other keys are correct. The outer radii of the annular domains are used as subordinate keys. Fig. 7c shows the decrypted image when the

incorrect subordinate keys are set as $r_1 = 36$, $r_2 = 72$, $r_3 = 108$, $r_4 = 144$, and $r_5 = 181$ while the other keys are correct. Using other random phases produced by computer to interfere the correct phases θ_i , $i = 1, 2, \dots, 5$, the decrypted image is shown in Fig. 7d. The correct phases ψ_i , $i = 1, 2, \dots, 5$ are interfered in the same way, and the decryption result is illustrated in Fig. 7e.

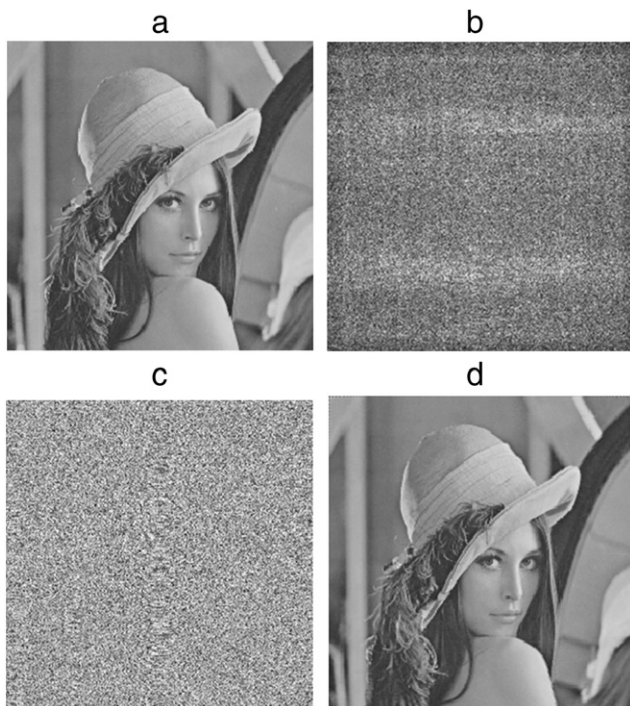


Fig. 6. (a) Original image; (b) the amplitude distribution of the encrypted image; (c) the phase distribution of the encrypted image; and (d) decrypted image.

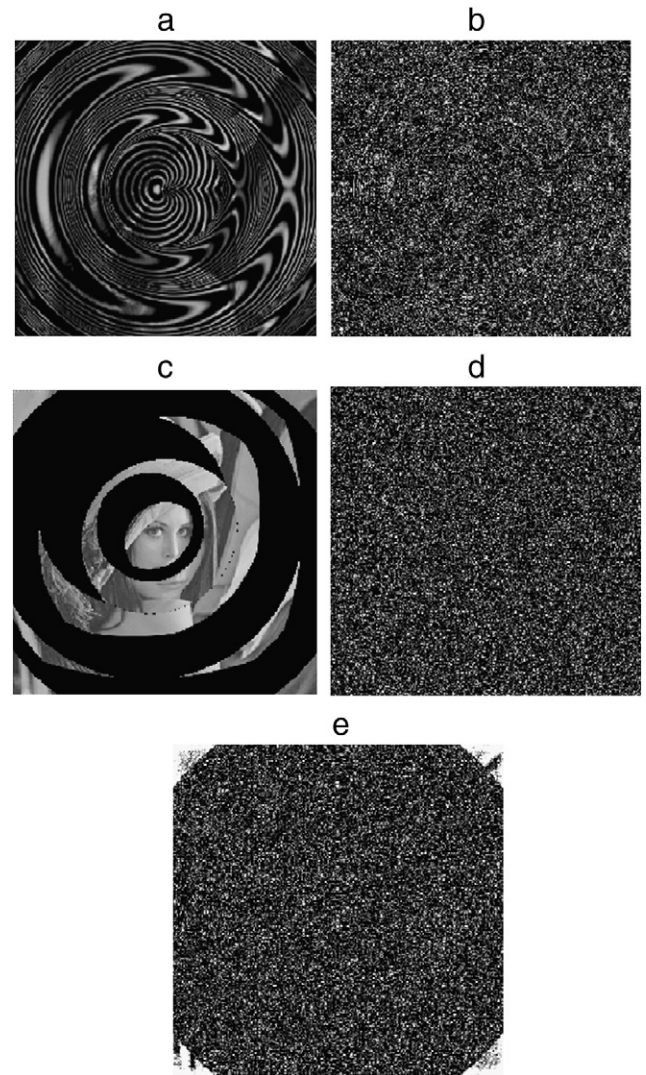


Fig. 7. Decrypted images with incorrect keys. (a) incorrect inverse FrMT order $p_i = 0.55$, $i = 1, 2, \dots, 5$; (b) incorrect inverse FrFT order $q = 0.6$; (c) incorrect outer radii of the annular domains set as $r_1 = 36$, $r_2 = 72$, $r_3 = 108$, $r_4 = 144$, and $r_5 = 181$; (d) incorrect phase θ_i , $i = 1, 2, \dots, 5$; and (e) incorrect phase ψ_i , $i = 1, 2, \dots, 5$.

5.1. Statistical analysis

Statistical analysis has been performed on the proposed image encryption algorithm, demonstrating its confusion and diffusion properties which strongly resist statistical attacks.

Image histogram is a very important feature in image analysis. Fig. 8a is the gray image Baboon which is obviously different from the image Lena shown in Fig. 6a. We can also learn it from the histograms of the two images shown in Fig. 8c and d, respectively. Fig. 8b gives the encrypted image of Baboon. Fig. 8e and f are the histograms for the encrypted Lena and Baboon, and they are quite similar. After a number of parallel experiments we can conclude that the ciphertext of different original images have similar histograms. So attackers cannot obtain useful information according to the statistical properties.

Entropy is a statistical measure of randomness that can be used to characterize the texture of an image. It is well known that the entropy $H(m)$ of a message source m can be calculated as

$$H(m) = - \sum_{i=0}^{2^L-1} p(m_i) \log_2 p(m_i), \quad (21)$$

where $p(m_i)$ represents the probability of message m_i , $L=8$. The entropy of the image Lena is 7.2157, while that of the encrypted Lena is 7.7155. The increase of entropy reflects that the distribution of gray-scale becomes more even.

Adjacent image pixels are highly correlated either in horizontal, vertical or diagonal directions. To test the correlation between two horizontal adjacent pixels, 2000 pairs of two horizontal adjacent

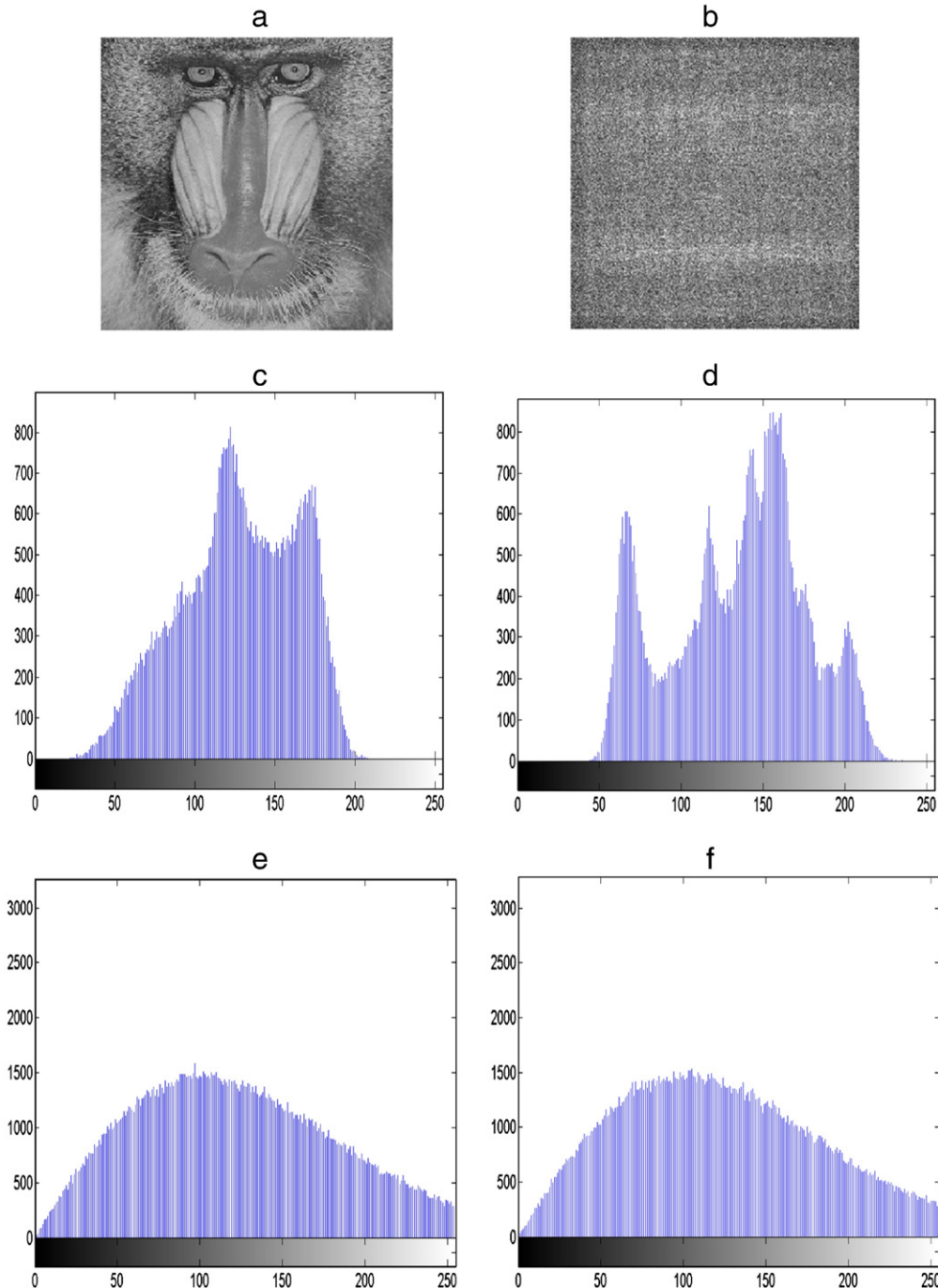


Fig. 8. (a) Image of Baboon; (b) encrypted image of Baboon; (c) histogram of Baboon; (d) histogram of Lena; (e) histogram of encrypted Baboon; and (f) histogram of encrypted Lena.

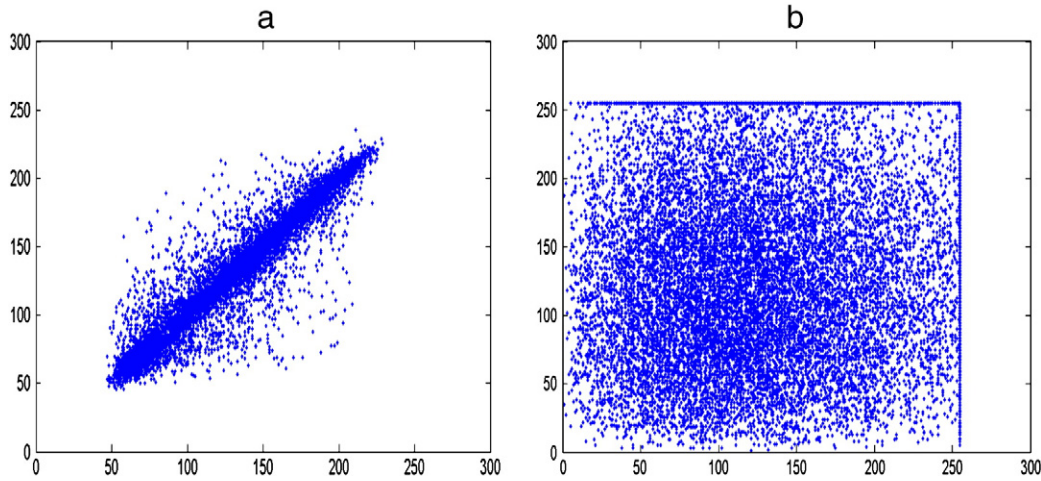


Fig. 9. Correlation distributions of two horizontally adjacent pixels (a) in the original image Lena and (b) in the encrypted image.

pixels are randomly selected from an image, and then the correlation coefficient can be obtained by

$$C = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\left(\sum_{i=1}^N (x_i - \bar{x})^2\right) \left(\sum_{i=1}^N (y_i - \bar{y})^2\right)}}, \quad (22)$$

where $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$ and $\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$. To test the correlation for two vertically adjacent pixels and two diagonally adjacent pixels, the same procedure is carried out. Fig. 9 gives the correlation distribution of two horizontally adjacent pixels in the original and the encrypted images of Lena. Table 1 gives the correlation coefficients of the Lena and encrypted Lena. From Table 1 we learn that the correlation of the encrypted image is much weaker than that of the original image. Consequently, the algorithm can be against statistical analysis.

In order to illustrate the randomness of the ciphertext, image Lena is encrypted with the FrMTs of orders p and $p + 0.1$ respectively while the other parameters are the same. The correlation coefficient between the two ciphertexts is 0.0219. After a number of parallel experiments, we learn that even if the same plaintexts are encrypted with different keys, their ciphertexts will be irrelevant. What's more, the ciphertext is completely different from the plaintext due to their different sizes, so we do not need to calculate the correlation between ciphertext and plaintext.

5.2. key sensitivity analysis

To express the quality of decrypted image, we introduce the logarithm of mean square error (LMSE) between decrypted image and the original image as

$$LMSE = \log_{10} \left\{ \frac{\sum_{x=1}^M \sum_{y=1}^N [f(x,y) - \bar{f}(x,y)]^2}{M \times N} \right\}, \quad (23)$$

Table 1
Correlation coefficients of adjacent pixels.

Correlation coefficient	Horizontal	Vertical	Diagonal
Original image	0.9613	0.9246	0.9246
Encrypted image	0.1089	0.0878	0.0683

where $f(x,y)$ and $\bar{f}(x,y)$ denote the pixel values of original image and decrypted image, respectively and $M \times N$ denotes the size of the image.

In the proposed image encryption algorithm, phase θ_i and ψ_i are main keys for their large key space. In order to reflect the result more clearly, we consider all phases $\theta_i, i = 1, 2, \dots, N$, are fluctuating in certain range, and the pseudo-keys θ'_i close to correct key are expressed as

$$\theta'_i = \theta_i + d\Delta\theta, \quad (24)$$

where $\Delta\theta$ is a random function whose value is located in the range $(-1, 1)$ and d is a coefficient. θ'_i is used for decrypting the image. The deviation of LMSE versus d is shown in Fig. 10a and the threshold of LMSE is 3.5, which is marked by a dash line. When the LMSE is less than the threshold, the partial or total information of original image can be recovered visually. The effective range of d is 2π for phase function and maximum value of $d\Delta$ is equal to 0.19. Consequently, the number of possible value for every point of key can be calculated by $2\pi/0.19 \approx 33$. And the result reflects a general effect on $N \theta_i$, i.e., $\theta_1, \dots, \theta_N$ (here $N=5$), therefore the number of possible value of entire keys is $33^{500 \times 500 \times 5} = 33^{500 \times 500 \times 5}$ which is a very huge number. ψ_i is analyzed in the same way, and its LMSE curve is shown in Fig. 10b. In Fig. 10b, $d\Delta$ is equal to 0.18, so the number of possible value of entire keys is $35^{500 \times 500 \times 5}$. It is generally intractable for cracker to carry out an exhaustive search of keys, thus the proposed algorithm is secure and resistant to brute-force attack [30].

Fig. 7a shows the decrypted image with incorrect FrMT orders and Fig. 11 shows the LMSE curve with the FrMT orders $p_i, i = 1, 2, \dots, N$. Here, all of the FrMT orders are equal and we consider all of the transform orders vary at the same time. The correct FrMT orders are equal to 0.5 in the encryption process, and different p_i in the range of $[0, 1]$ are tried to decrypt the image. The change of curve is not so violent, so the FrMT orders in the proposed algorithm are not so sensitive, but FrMT still plays a vital role in the encryption system. Fig. 12 gives the LMSE curve with the FrFT order q and Fig. 7b gives the decrypted image with incorrect FrFT order. The correct FrFT order q is equal to 0.5, and we assume that other keys in the decryption process are correct. It can be seen that the decrypted image has very distinct difference even if a very small error occurs in the fractional order q . When the deviation of decryption order is up to 0.005, one has difficulty in recovering the original image from the encrypted image. Thus, the FrFT order is a key with high sensitivity and security.

As we can see from the simulation results, the sensitivity of the FrMT order is relatively weaker than that of the FrFT. It is known that when a slight deviation occurred to the FrFT order, high-frequency components of the original image cannot be recovered, so the detail parts of the recovered image are blurry while the general parts of the

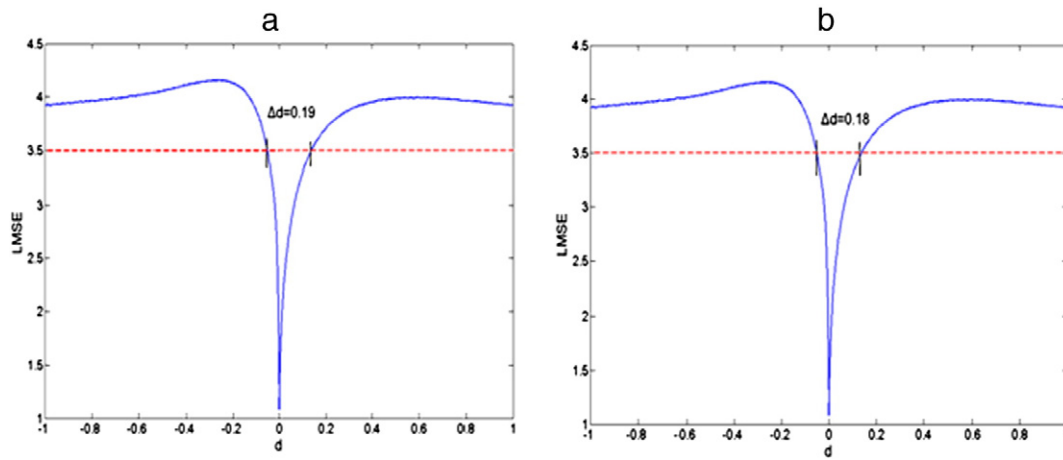


Fig. 10. LMSE curves for (a) θ_i and (b) ψ_i .

image can be recognized. But in the proposed encryption algorithm, when the image is transformed by FrMT, some parts of the image close to the transform center are magnified due to the logarithm operation. Thus the detail parts of the original image are not detail anymore here, and when a tiny deviation occurred to the FrMT order, the variation of the recovered image is not so prominent. Consequently, the order of FrMT is not as sensitive as that of FrFT.

Fig. 7c is the decrypted image with incorrect outer radii of the annular domains. One can see that some parts of the original image cannot be recovered and the recovered parts are twist and deformed to a certain extent. Fig. 13 gives the LMSE between the decrypted image and the original one varying with the number of incorrect outer radii of the annular domains. The more outer radii are incorrect, the more remarkable the differences between the decrypted images and the original image are, since the decrypted images become more anamorphic with the increase of incorrect outer radii. But the outer radii of the annular domains are not so sensitive, which just can be subordinate keys.

5.3. Security analysis

Common attacks, such as known-plaintext [17], chosen-plaintext [17], ciphertext-only [31] and chosen-ciphertext [32] attacks are usually used to test the security of the image encryption algorithm. To simplify the analysis, we assume that all data is available in digital form. As for a real optical implementation, the difficulty to manipulate and measure the involved physical quantities adds some challenges to the attacker.

Known-plaintext, chosen-plaintext and chosen-ciphertext attacks require some known extra information, such as plaintext–ciphertext pairs, and some of them even require that the plaintext or the ciphertext has very specific form which is chosen by the attacker. In a linear

encryption scheme, if P_1, P_2 are two plain images with corresponding ciphered image C_1, C_2 , and α, β are two scalar coefficients, it is easy to see that $\alpha P_1 + \beta P_2$ is encrypted to $\alpha C_1 + \beta C_2$. According to above relationship, a particular linear combination can give a delta function to attack the encryption system, or linear equations $\mathbf{C} = \mathbf{TP}$ [17] which characterize the encryption process can be constructed to attack the encryption system. In our proposed algorithm, the above relationship is invalid due to the nonlinear property of the FrMT. **Even if a Dirac delta function can be used as plain image directly, these attacks are useless not only because there are more than one pixel in ciphertext corresponding to one pixel in plaintext after the FrMT but also because it is difficult and impractical to extract such large number of keys. So we just highlight ciphertext-only attack here.**

The known data are $C(x, y)$ and all of the keys of the encryption system except phases θ_i and ψ_i ($i = 1, 2, \dots, N$, here $N = 5$). A scheme of ciphertext-only attack can be designed based on the phase retrieval algorithm which is called Gerchberg–Saxton (GS) algorithm. The GS phase retrieval technique is applied to access the encryption keys θ_i and ψ_i . First, unknown phases θ_i and ψ_i are replaced with random phases generated by computer and they are used as secret keys to decrypt the ciphertext. Then decrypted image is used as original image to access encryption process, and a series of keys can be retrieved. These retrieved keys substitute for θ_i, ψ_i and above process is repeated. After several iterations, we obtain some equivalent keys, and they are used to decrypt ciphertext. If the original image can be recovered, the encryption algorithm fails in ciphertext-only attack, otherwise the algorithm against ciphertext-only attack. LMSEs between the original and each iterative retrieved image versus the iterative number are calculated as shown in Fig. 14b. It is obvious that the original image cannot be recovered at all, so the proposed encryption algorithm is secure against the ciphertext-only attack.

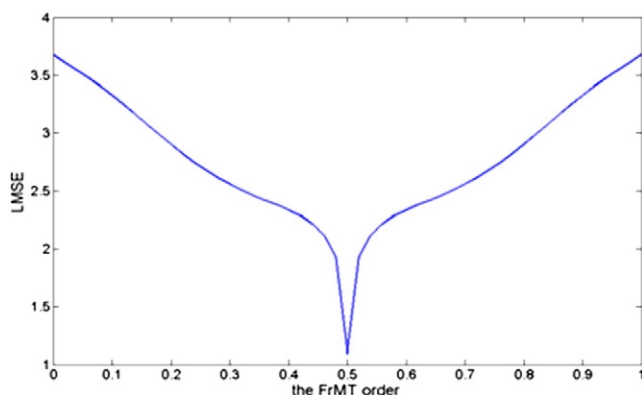


Fig. 11. LMSE curve for the FrMT order.

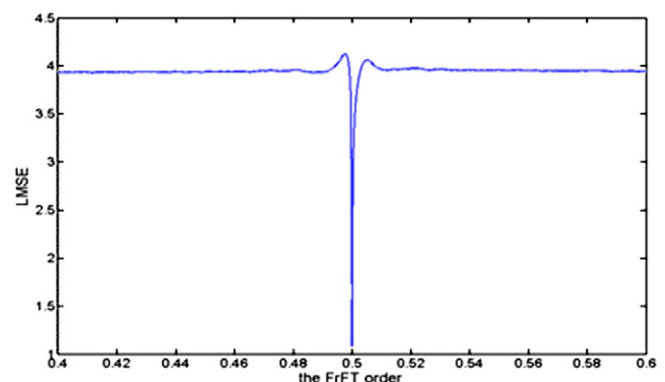


Fig. 12. LMSE curve for the FrFT order.

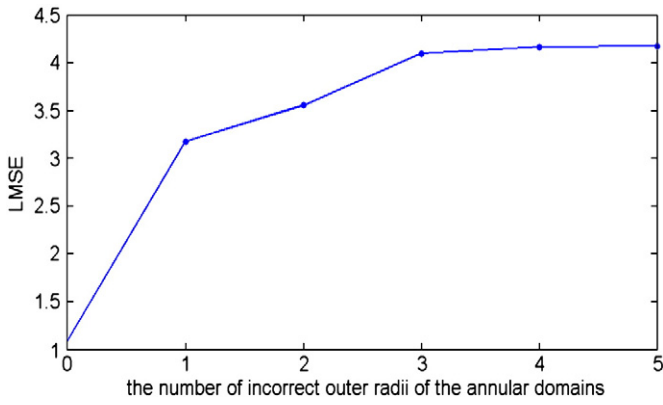


Fig. 13. LMSE curve for the number of incorrect outer radii of the annular domains.

5.4. Robustness analysis

It is inevitable that the noise impact directly on the quality of the decrypted image at the stage of image processing and image transmission, thus we have to study the performance of the proposed scheme against the noise attack. For the noise attack, the noise is added into the encrypted image in the following way

$$C' = C(1 + kG), \quad (25)$$

where C and C' are the encrypted image and noisy encrypted image respectively, k is a coefficient indicating the noise strength, and G represents Gaussian random data with zero-mean and the standard deviation of 1. The mean square error (MSE) can be calculated by Eq. (23) without taking logarithm. The Gaussian noises with different

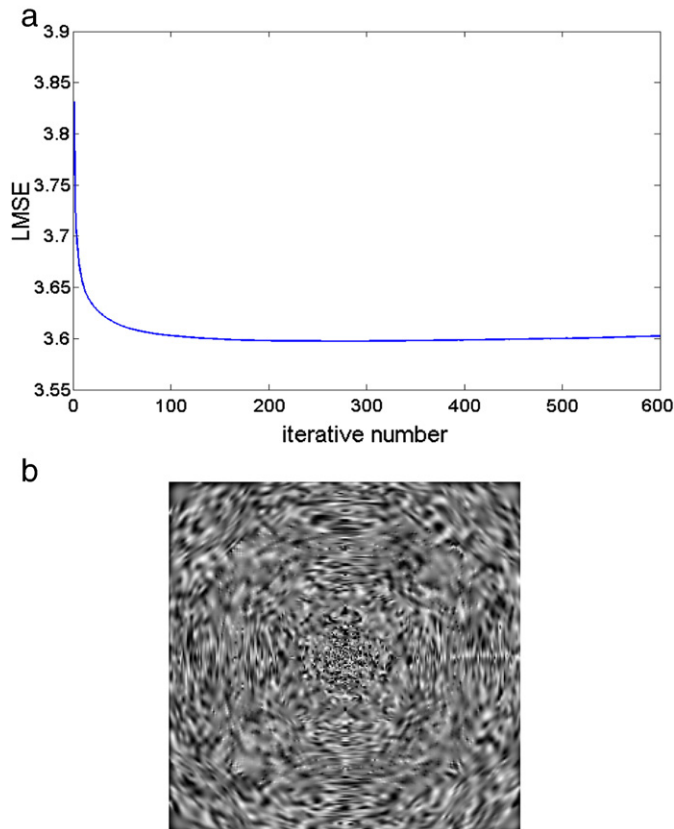


Fig. 14. The results of ciphertext-only attack: (a) LMSE curve, and (b) the decryption image with 600 iterations.

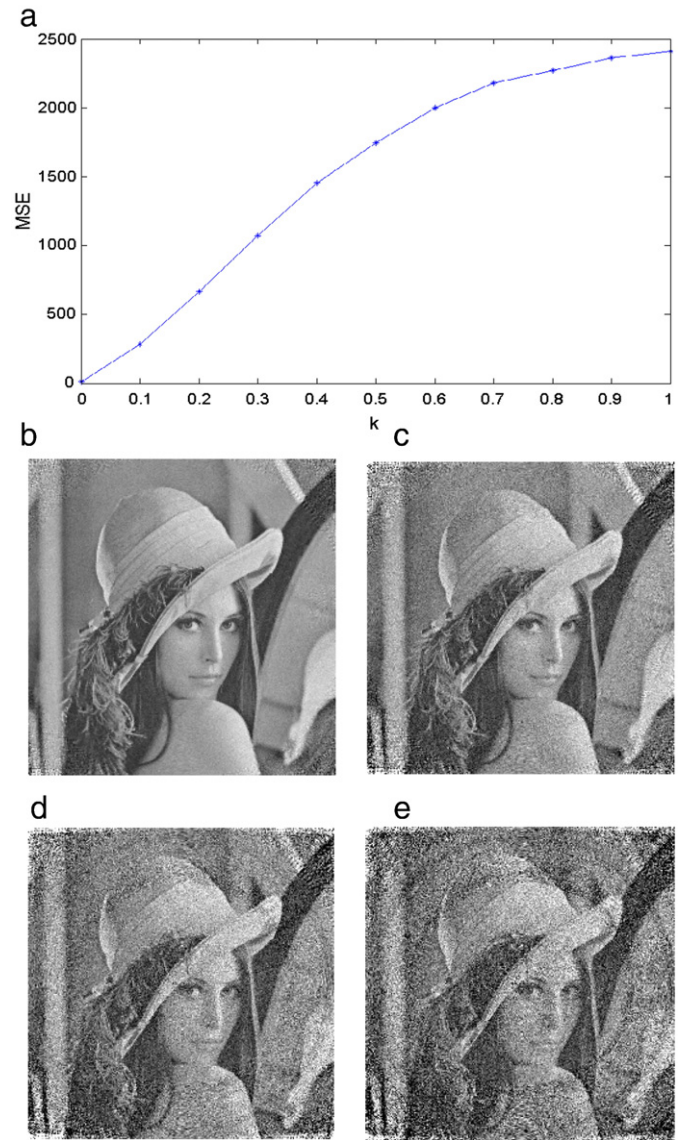


Fig. 15. Results of noise attack: (a) The MSE curve, (b) $k = 0.05$, (c) $k = 0.2$, (d) $k = 0.5$, and (e) $k = 1$.

noise intensities are added to the encrypted image, and the deviation of MSE versus noise intensity k is illustrated in Fig. 15a. Fig. 15b–e shows the decrypted images when k equals to 0.05, 0.2, 0.5 and 1. As one can see from Fig. 15, the decrypted images can still be recognized within a certain range of noise level. Although the decrypted image become blurrier with the increase of noise intensity, but the major information of the image can still be obtained even when $k = 1$.

The performance of the proposed technique is analyzed against the occlusion of the encrypted data. The encrypted images with 1/8, 1/4 and 1/2 occlusion are shown in Fig. 16a, c and e, and the corresponding decrypted images are shown in Fig. 16b, d and f, respectively. Though the decrypted images are rough versions of the original image, the main information of the original image can be known with correct keys.

6. Conclusion

We proposed an optical image encryption algorithm based on fractional Mellin transform. FrMT makes the proposed scheme nonlinear and less vulnerable to attacks compared with the conventional linear encryption techniques. Different annual domains

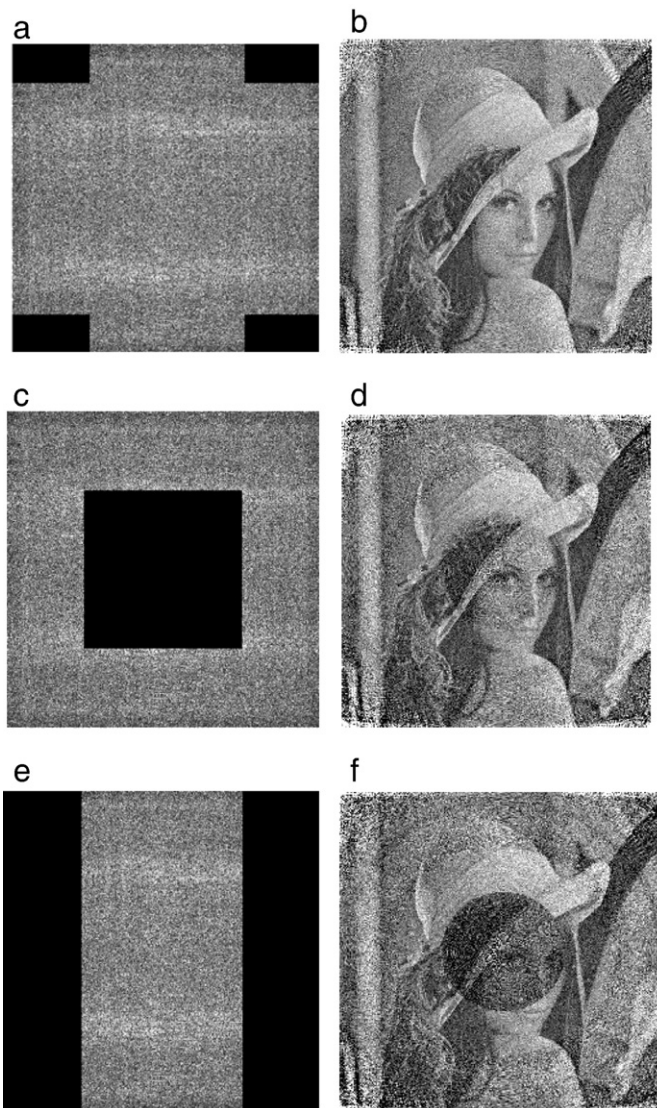


Fig. 16. Results for occlusion of encrypted data. The encrypted images with (a) 1/8, (c) 1/4 and (e) 1/2 occlusion. Decrypted images for (b) 1/8 occlusion, (d) 1/4 occlusion and (f) 1/2 occlusion of the encrypted data, respectively.

of the original image are transformed by FrMTs of different orders. The outputs are some complex-valued images, and they are further encrypted through iterations based on amplitude encoding and phase encoding. The main keys are some phases generated in the iterative process. Moreover, the FrMT orders, the outer radii of different annular domains and the FrFT order are also regarded as keys

optionally. So the key space of the proposed algorithm is very large. Although the orders of FrMTs are not so sensitive, it doesn't impact the performance of the proposed encryption system, since the primary purpose of employing the FrMT is to strengthen the capability of the encryption system against common attacks and other keys of the proposed encryption technique are sensitive enough to satisfy the sensitivity requirement for an encryption system. The optoelectronic hybrid setup of the proposed encryption algorithm has been given. Simulation results show that the proposed system is secure against the statistic attack, brute force attack and ciphertext-only attack. Except its high security, the proposed algorithm is demonstrated to have good robustness as well.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (grant no. 10647133), the Natural Science Foundation of Jiangxi Province (grant nos. 2009GQS0080 and 2007GQS1906), and the Scientific Research Start-up Funds for the Recruitment of Talent of Nanchang University.

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